

Vinal DalcY Dsouza

Data Engineer | AI Researcher

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PROJECTS

AI PDF Assistant [FastAPI, pgvector, PostgreSQL, RRF, Streamlit, PyTorch, Docker, Alembic, pytest]

Jan 2026 – Apr 2026

- Cut hallucinations without sacrificing recall, by **fusing keyword and semantic search** (RRF), **citing sources** on every answer, and short-circuiting to "I don't know" below a confidence threshold.
- Built a **custom LLM-as-judge** (Groq Llama-70B) with full Langfuse tracing, giving every change a live readout on recall, precision, and hallucination rate before it shipped.
- Hardened against real attack vectors, by layering **SSRF defenses** (IP blocklists, redirect re-validation) and **gating CI** on mypy – strict, ruff and CVE scanning.

Project 42, A Terraform MCP [Python, MCP, GraphQL, LLM, Claude Agent SDK, Claude Code, Cursor]

Jan 2026 – Apr 2026

- Turned an ambiguous problem, "how should an AI agent reason about live infrastructure", into a discrete, testable interface, by designing 17 purpose-built MCP tools with a GraphQL query layer over Terraform state.
- Cut infrastructure dependency-query costs by **10x**, by replacing raw-state access with structured, scoped retrieval that agents could reliably act on.

EXPERIENCE

Graduate Research Assistant

Sep 2025 – May 2026

Northeastern University, Accepted at PEPR'26.

Oakland, CA

- Ran a structured **inter-rater reliability study** using Cohen's Kappa and Likert scaling to calibrate human judgments of AI output across 45 government employees, diagnosing adoption gaps by embedding directly with **14+** municipal departments during a live Copilot deployment.

Senior Data Engineer

Mar 2022 – Jun 2024

Accenture —UK based Financial Services Client

India

- Took a proof of concept to full production adoption, cutting processing time from **6 hours to 36 minutes** on 5M+ records, validated at 100% accuracy over a 30-day parallel run before cutover.
- **Built a meta data driven ingestion accelerator** adopted by multiple delivery teams, reducing new-source onboarding from **1 week to 2 days**, by inferring schema and generating bronze tables automatically from connector details alone.
- Automated manual handoffs across critical credit-risk milestones with failure detection and ownership routing, **cutting support escalations 50%**.
- Sped up a regulatory financial report from **30 minutes to 15 seconds** by rewriting slow queries and improving data indexing.
- Designed Databricks ingestion pipelines handling Kafka streams and multi-format batch feeds (XML, JSON, Oracle, flat files), implementing idempotent incremental loads, schema validation, and recovery workflows to maintain **99.9% SLA**.

Senior Systems Engineer

Sep 2019 – Mar 2022

Infosys —US based Banking & Capital Markets

India

- **Optimized Public Assets warehouse** holding hundreds of millions of records (CUSIP, MLSEC, ISIN etc), accelerating execution plans by **40%** on complex queries fetching active asset states to ensure timely SFTP delivery downstream.
- Reduced extract onboarding from **2 days to 3 hours** by building a **metadata-driven framework** where an Oracle config table mapped extract IDs to SQL logic, enabling a single DataStage job to generate any new deliverable with no custom pipeline.
- **Built a Jenkins CI/CD pipeline for DataStage**, automating DSX packaging and SAT/UAT deployments to improve release reliability and traceability.

SKILLS

AI & Agents: LLM APIs (OpenAI, Anthropic, Groq), MCP, Agent SDKs, Agentic Workflows, RAG, Prompt Engineering, Evals (Ragas), LangChain, Langfuse, Slack Integration, n8n, Pytorch, Scikit-learn

Languages: Python, SQL, JavaScript, Node.js, React

Data & Cloud: PySpark, Databricks, AWS (S3, Lambda, EC2, IAM, SageMaker), Kafka

Infra & Devops: Docker, Terraform, GitHub Actions, CI/CD, Jenkins, Oracle

EDUCATION

M.S. Computer Science | Northeastern University, Oakland, CA | GPA 4.0 / 4.00

Sep 2024 – May 2026

B.E. Information Science | NMAM Institute of Technology, Nitte, India | GPA 9.09 / 10.0

Aug 2015 – Jun 2019

Certifications: Databricks Fundamentals • AWS Academy Cloud Architecting • AWS Academy Machine Learning Foundations